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La traslación converge en L^p a la función original

Lema 1. *Sea* $p \in [1, \infty)$ *y* $f \in \mathcal{L}^p(\mathbb{R}^n)$, *entonces* $T_h f \xrightarrow[h \to 0]{\mathcal{L}^p} f$.

***Demostración*:** Como $\mathcal{C}_c(\mathbb{R}^n)$ es denso en $L^p(\mathbb{R}^n)$ para $p \in [1, \infty)$ (`teo-fn-continua-soporte-compac` dado $\varepsilon > 0$, existe $g \in \mathcal{C}_c(\mathbb{R}^n) : \|f - g\|_p < \frac{\varepsilon}{3}$. Como $g \in \mathcal{C}_c(\mathbb{R}^n)$, $|g|^p \in \mathcal{L}^1(\mathbb{R}^n)$ y $\lim_{h \to 0} g(x - h) = g(x)$.

$$\implies \lim_{h \to 0} \|T_h g - g\|_p^p = \lim_{h \to 0} \int_{\mathbb{R}^n} |g(x - h) - g(x)|^p \, \mathrm{d}x \overset{\substack{\text{TCD} \\ \downarrow}}{=} \int_{\mathbb{R}^n} \lim_{h \to 0} |g(x - h) - g(x)|^p \, \mathrm{d}x = 0.$$

Entonces, $\exists \delta > 0 : \forall h \in B(0, \delta) : \|T_h g - g\|_p < \frac{\varepsilon}{3}$. Por la desigualdad de Minkowski,

$$\begin{aligned} \|T_h f - f\|_p &\leqslant \|T_h f - T_h g\|_p + \|T_h g - g\|_p + \|g - f\|_p \\ &\leqslant \|T_h(f - g)\|_p + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} \leqslant \varepsilon. \end{aligned}$$

Haciendo $\varepsilon \to 0$, $\lim_{h \to 0} \|T_h f - f\|_p = 0$. ■

Referenciado en

- `teo-aproximacion-identidad-convolucion`
- `lem-riemann-lebesgue-l1`
- `prop-convolucion-exp-conjugados`
- `lem-riemann-lebesgue-r`

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